Time as Projection: The Klein-Gordon Spectrum on $S^3 \times \mathbb{R}$ and the Algebraic / Observation Split

GeoVac Project

May 2026

Abstract

The bare relativistic Klein-Gordon spectrum on $S^3 \times \mathbb{R}$ is π-free in the algebraic-extension ring $\mathbb{Q}[\sqrt{d_1}, \sqrt{d_2}, \ldots]$ with d_i positive square-free integers, for every rational mass-squared parameter. We verify this symbolically for $n \in [1, 50]$ across the panel $m^2 \in \{0, 1, 1/4, 2\}$ (200 cases, zero transcendentals). Mass is therefore a parameter projection that preserves the bare-graph algebraic ring; it does not introduce π.

π enters the spectrum at exactly one step: temporal compactification. On $S^3 \times S^1_\beta$ the temporal direction acquires Matsubara modes $\omega^t_k = 2\pi k/\beta$, and the first π-bearing eigenvalue is the $(n{=}0, k{=}1)$ mode with $\omega^2 = 4\pi^2$. We exhibit the before/after explicitly: every $(n, 0)$ eigenvalue is π-free; every $(n, k{\neq}0)$ eigenvalue contains $2\pi/\beta$.

The conformally coupled massless scalar Casimir energy on unit S^3 (spatial only, no temporal compactification) evaluates to the exact rational $E_{\text{Cas}} = 1/240$, with no transcendental content at any step of the ζ-regularized derivation. The Stefan-Boltzmann constant $\pi^2/90$ appears in the high-temperature expansion of the free-energy density on $S^3 \times S^1_\beta$ only via the Matsubara sum, exactly through the mechanism of the spectral π-injection.

We propose a refinement of Paper 34's projection dictionary: the implicit "scalar parameter projection" should be split into (i) a *rest-mass projection* (introduces m, dimension mass, transcendental signature trivial, ring-preserving for $m^2 \in \mathbb{Q}$) and (ii) an *observation / temporal-window projection* (introduces β, dimension time, transcendental signature $2\pi \cdot \mathbb{Q}$ per Matsubara mode; Stefan-Boltzmann $\pi^2/90$ in the high-T limit). These are categorically distinct; lumping them as a single "parameter row" under-resolves the transcendental axis of the dictionary.

The structural reading is sharper than Paper 34's layer description: *a GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter that has been promoted from the discrete graph spectrum.* The discrete graph itself is π-free; the projection that integrates is where π appears. Existing GeoVac observables (Pauli counts, magic numbers, hydrogenic eigenvalues, Bargmann–Segal lattice, graph-native QED) are confirmatory; the canonical π-bearing observables (vacuum-polarization $\Pi = 1/(48\pi^2)$, β-function $2\alpha^2/(3\pi)$, Stefan-Boltzmann) are all evaluated through continuous integrations of the temporal kind.

Beyond the verification sprints KG-1, KG-2, KG-3 and the closed forms reported in Sections 5 and 7.3 (among them the S^5 conformal $-31/60480$ and the Bargmann–Segal harmonic-oscillator $-17/3840$, both computed here), this paper introduces no new computation. Its contribution is the structural observation that *time, not mass*, is the projection axis along which π enters the framework, and that the observer's finite-time measurement window is the formal mechanism.

1 Introduction

GeoVac Paper 34 [12] introduced a two-layer reading of the framework: Layer 1 is a bare combinatorial graph (quantum-number labels, integer eigenvalues, rational matrix elements, no physics), and Layer 2 is a finite family of named projections from the bare graph to physical settings. Each projection introduces specific physical variables, specific dimensions, and a specific transcendental signature. Paper 18 [4] provides the parallel transcendental classification: intrinsic, calibration, embedding, flow, composition. Paper 24 [7] establishes the Bargmann–Segal lattice as bit-exactly π-free; Paper 28 [8] establishes graph-native QED as algebraic in $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$.

The natural question, which Paper 34 leaves open and Paper 31 [9] sharpens through the universal/Coulomb partition, is: where exactly does π enter the framework, and which projections in the Layer 2 dictionary are responsible? Most of the canonical π-bearing observables in GeoVac to date (vacuum polarization, β-functions, Stefan–Boltzmann constants, fine-structure calibration $1/(4\pi)$) have been pinned to specific projections empirically – the Hopf bundle for the α Observation, the Connes–Chamseddine spectral action for one-loop QED, the vector-photon promotion for the $1/(4\pi)$ per-loop Weyl measure on the Hopf base. But there is no single statement of the form "π enters at step X, and at no other step."

This paper makes such a statement, in the context of the simplest non-trivial relativistic system the framework can describe: a free scalar field on $S^3 \times \mathbb{R}$, governed by the Klein-Gordon equation. The Klein-Gordon spectrum is well-known and standard; the contribution here is that, when computed within the GeoVac algebraic-ring tracking discipline, it provides a clean and mechanically verifiable "before / after" for the appearance of π.

The result has a direct consequence for Paper 34's projection dictionary: the row that has been implicitly conflating "introduces a mass" with "introduces an observation window" is under-resolved. These are categorically distinct projections on the transcendental axis. The first is ring-preserving; the second injects 2π.

Sprint provenance

The verification computations (KG-1, KG-2, KG-3)[1] are documented in `debug/kg_sprint_memo.md` and the three companion JSON data files; the scripts that produced them are `debug/kg1_algebraic_ring.py`, `debug/kg2_temporal_compactification.py`, and `debug/kg3_casimir_s3_s1.py`. All three completed without blocked items; one minor implementation bug (`sp.nsimplify` mis-rewriting rationals) was identified and fixed during KG-1 and is documented in the memo.

2 Klein-Gordon on $S^3 \times \mathbb{R}$: setup

Take the metric $ds^2 = -dt^2 + R^2 d\Omega_3^2$ where $d\Omega_3^2$ is the round metric on the unit S^3 and R is the spatial radius. The Klein-Gordon equation

$$(\Box + m^2)\phi = 0 \tag{1}$$

separates as $\phi(t, \Omega) = e^{-i\omega t} Y_n(\Omega)$, with Y_n the hyperspherical harmonics.

The scalar Laplace–Beltrami on the unit S^3 has eigenvalues

$$-\Delta_{S^3} Y_n = n(n+2)\, Y_n, \quad n = 0, 1, 2, \ldots, \tag{2}$$

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

with degeneracies $(n+1)^2$. This is standard SO(4) representation theory; the eigenvalues are integers, the multiplicities are perfect squares, and no transcendental content appears. Note that this is the bare scalar Laplacian; it is distinct from the Fock-projected Coulomb operator with eigenvalues $\lambda_n = -(n^2-1)$ used elsewhere in the framework [2].

The dispersion relation from (1) and (2) is

$$\omega_n^2 = \frac{n(n+2)}{R^2} + m^2. \tag{3}$$

Take $R = 1$ throughout (units restored only at the level of physical observables); then $\omega_n = \sqrt{n(n+2)+m^2}$.

3 The algebraic ring (KG-1)

Observation 1 (KG-1: π-free certificate for the bare KG spectrum)**.** *For every $n \in [1, 50]$ and every $m^2 \in \{0, 1, 1/4, 2\}$, the Klein-Gordon eigenvalue $\omega_n = \sqrt{n(n+2)+m^2}$ admits a unique decomposition*

$$\omega_n = c_n \cdot \sqrt{d_n}, \quad c_n \in \mathbb{Q}, \quad d_n \in \mathbb{Z}_{>0} \text{ square-free.} \tag{4}$$

Symbolic verification (`sp.simplify`$(\omega_n^2 - c_n^2 d_n) = 0$*) passes for all 200 cases. No transcendentals (π, e, log, ζ-values) appear in any ω_n for any rational m^2 in the panel.*

The square-free generators that appear vary by mass. A representative slice (first six values of n):

m^2	ω_1	ω_2	ω_3	ω_4
0	$\sqrt{3}$	$2\sqrt{2}$	$\sqrt{15}$	$2\sqrt{6}$
1	2	3	4	5
1/4	$\sqrt{13}/2$	$\sqrt{33}/2$	$\sqrt{61}/2$	$\sqrt{97}/2$
2	$\sqrt{5}$	$\sqrt{10}$	$\sqrt{17}$	$\sqrt{26}$

The $m^2 = 1$ case is structurally special: $n(n+2)+1 = (n+1)^2$, so the spectrum collapses to the integers $\omega_n = n+1$. This is exactly the conformally coupled massless scalar on the unit S^3 — the conformal mass shift ξR at $\xi = 1/6$ on a manifold of constant scalar curvature $R = 6$ (the unit S^3) gives an effective $m^2_{\text{eff}} = 1$ [14]. The collapse is what makes the spectral zeta of the conformally coupled scalar reduce to the ordinary Riemann zeta (Section 5).

For $m^2 \in \{0, 1/4, 2\}$, the union of square-free generators that appear in $n \in [1, 50]$ contains 136 distinct integers ranging from 2 to 10401, all in $\mathbb{Q}[\sqrt{d}]$ for some d. (Recounted 2026-08-28; the previously quoted "134…from 1" was wrong in both the count and the lower endpoint — 1 is not a generator in this set.)

Negative controls. For $m^2 = 1/\pi$ and $m^2 = \sqrt{2}$ (both irrational), the ring is broken at $n = 1$ and the corresponding irrational propagates into ω_n^2 for all n. The mechanism is trivial – the spectrum is a literal sum – but the observation pins the boundary: the Klein-Gordon eigenvalue is in $\mathbb{Q}[\sqrt{d_1}, \sqrt{d_2}, \ldots]$ if and only if m^2 is in the same ring. The mass parameter is, by itself, transcendental-free *when supplied as a rational number*; the framework does not introduce additional π during the parameter projection.

Remark 1 (Comparison with Paper 24 and Paper 28)**.** *Paper 24's π-free certificate for the Bargmann–Segal S^5 HO graph and Paper 28's algebraic-extension certificate for graph-native QED were both established at the level of matrix elements of the discrete graph operators. Observation 1 extends the same kind of certificate to the relativistic spectrum on $S^3 \times \mathbb{R}$, treating the spectrum itself (not just the graph operator) as the object certified. The three certificates partition cleanly: Paper 24 covers the bare HO graph adjacency, Paper 28 covers the finite-graph QED matrix elements, and the present observation covers the relativistic spectrum of a free field on $S^3 \times \mathbb{R}$. None of them require π.*

4 Where 2π enters (KG-2)

The Klein-Gordon equation on $S^3 \times \mathbb{R}$ has the π-free spectrum of Section 3. The Klein-Gordon equation on $S^3 \times S^1_\beta$, where the temporal direction is periodically identified with period β, has the spectrum

$$\omega_{n,k}^2 = n(n+2) + \left(\frac{2\pi k}{\beta}\right)^2 + m^2, \quad n = 0, 1, 2, \ldots, \quad k \in \mathbb{Z}. \tag{5}$$

The temporal eigenvalues $\omega_k^t = 2\pi k/\beta$ are the standard Matsubara modes; their factor of 2π is the ratio of one full circle (2π in radian measure) to the circumference β of the identified temporal direction. This factor is structurally identical to the $1/(4\pi)$ that appears in the Hopf-base Weyl measure [11]: it is the calibration constant attached to the compact-direction integral measure.

The before/after is mechanical:

Step	Eigenvalue object	Transcendentals
Spatial Laplace–Beltrami on S^3	$-\Delta Y_n = n(n+2)Y_n$	none (integers)
KG dispersion on $S^3 \times \mathbb{R}$, t open	$\omega_n^2 = n(n+2) + m^2$	none (rational m^2)
Periodic time, $k = 0$ sector	$\omega_{n,0}^2 = n(n+2) + m^2$	none
Periodic time, $k \neq 0$ sector	$\omega_{n,k}^2 = n(n+2) + (2\pi k/\beta)^2 + m^2$	$\boldsymbol{2\pi}$ *appears here*

Observation 2 (KG-2: temporal compactification injects 2π)**.** *The first eigenvalue of* (5) *that contains π is the* (n=0, k=1) *Matsubara mode at $\omega^2 = 4\pi^2/\beta^2$. At $\beta = 1$, $m = 0$, the first seven eigenvalues sorted by ascending ω^2 are (listing ω): 0, $\sqrt{3}$, $2\sqrt{2}$, $\sqrt{15}$, $2\sqrt{6}$, $\sqrt{35}$, 2π. The first six are in $\mathbb{Q}[\sqrt{d}]$; the seventh is the first π-bearing one. (Corrected 2026-08-28: an earlier version listed five and placed 2π fifth, omitting the $n = 4, 5$ spatial modes $2\sqrt{6} \approx 4.899$ and $\sqrt{35} \approx 5.916$, both below $2\pi \approx 6.283$. The load-bearing claim — that 2π is the* first *π-bearing eigenvalue — is unaffected.) No earlier step in the construction injects π.*

The "injection" is structurally the same statement as the Peter–Weyl realization on S^1: the compact circle of length β has Laplacian eigenvalues $(2\pi k/\beta)^2$ because the smallest non-trivial wavenumber on a circle of length β is $2\pi/\beta$. There is no other source. All higher π^k content in the finite-temperature partition function, free energy, and Casimir energy on $S^3 \times S^1_\beta$ traces back to multiple Matsubara modes multiplying.

5 Casimir energy on S^3: a one-projection match (KG-3)

The Casimir energy of a conformally coupled massless scalar on the unit S^3 (no temporal compactification: temperature $T = 0$) is zeta-regularized as

$$E_{\text{Cas}} = \frac{1}{2}\zeta_X(-1), \quad \zeta_X(s) = \sum_{n=0}^{\infty}(n+1)^2\,\omega_n^{-s}, \tag{6}$$

where the spectrum and degeneracy are those of the conformally coupled massless scalar. By the integer collapse of Section 3, $\omega_n = n + 1$ and the spectral zeta reduces to the Riemann zeta shifted by two arguments:

$$\zeta_X(s) = \sum_{n=0}^{\infty}(n+1)^{2-s} = \zeta_R(s-2). \tag{7}$$

Evaluating at $s = -1$:

$$E_{\text{Cas}} = \frac{1}{2}\zeta_R(-3) = \frac{1}{2}\cdot\frac{1}{120} = \boxed{\frac{1}{240}} \tag{8}$$

where $\zeta_R(-3) = -B_4/4 = 1/120$ via the Bernoulli number identity ($B_4 = -1/30$). The result is exact rational, with no transcendental content at any step.

Observation 3 (KG-3: spatial Casimir on S^3 is π-free)**.** *The ζ-regularized Casimir energy of the conformally coupled massless scalar field on the unit S^3 is the exact rational $E_{Cas} = 1/240$. The full per-step transcendental ledger is:*

Step	*Quantity*	*Transcendentals*
Spatial spectrum (conformal)	$\omega_n = n+1$, *deg.* $(n+1)^2$	*none*
Spectral zeta	$\zeta_X(s) = \zeta_R(s-2)$	*none*
Evaluate $\zeta_R(-3)$	$-B_4/4 = 1/120$	*none*
Casimir	$E_{Cas} = 1/240$	*none*

The match against the textbook value (Bytsenko et al. [15], Dowker–Critchley [16], Ford [17]) is exact rational equality, verified to 60 decimal places numerically.

The structural significance of Observation 3 is that the spatial Casimir on S^3 provides Paper 34 with a clean, exactly computable, single-projection match: the projection invoked is the Fock conformal projection (Paper 34 Table 1 row 1) plus the conformal coupling shift; the transcendental signature is rational; and the value 1/240 matches an independently established physics quantity. By the Layer-2-presence bound of Paper 34 (this is a zero-Layer-2-input, rational-analytic chain), a single-projection match of this kind should sit at the framework's basis-quality limit – here exact – and indeed it is.

Observation 4 (The dimensional Casimir ladder: S^1, S^3, S^5 and the Bernoulli rungs (Sprint GB-3))**.** *The conformal-scalar Casimir on S^{2k-1}, $E_{Cas} = \frac{1}{2}\sum_l \omega_l \deg_l$ with $\omega_l = l + (d-1)/2$, leads with $\zeta_R(-(2k-1))$, i.e. the Bernoulli number B_{2k}. The first three rungs:*

Manifold	$\omega\cdot\deg$ *(in n)*	E_{Cas}	*rung*
S^1	n	$\frac{1}{2}\zeta(-1) = -1/24$	B_2
S^3	n^3	$\frac{1}{2}\zeta(-3) = 1/240$	B_4
S^5	$(n^5-n^3)/12$	$\frac{1}{24}(\zeta(-5)-\zeta(-3)) = -31/60480$	B_6 *(+* B_4*)*

On S^3 the degeneracy n^2 keeps $\omega \cdot \deg$ a single power (n^3), so the Casimir is a clean single rung. On S^5 the quartic degeneracy $n^2(n^2-1)/12$ spreads $\omega \cdot \deg$ across n^5 and n^3, so the S^5 Casimir carries both *B_6 ($\zeta(-5) = -1/252$, coefficient $+1/24$) and B_4 ($-1/24$), with B_2* absent. *The value $-31/60480$ is exact (verified in `sympy`) and, to our knowledge, newly computed; S^5 is a GeoVac manifold via the Bargmann–Segal lattice (Paper 24). The spreading is the same multiplicity mechanism that obstructs the Dirac-zeta functional equation (Paper 28's functional-equation obstruction), and the S^3/S^5 rungs place the framework's Casimirs on the Bernoulli ladder of Paper 32 (the two-layer split read as the ζ functional-equation reflection $\zeta(s) \leftrightarrow \zeta(1-s)$, with the transcendental window $\zeta(2k) = \text{rational} \cdot \pi^{2k}$ supplying the M2 observables and the negative-integer window $\zeta(1-2k)$ the π-free skeleton). Source: `debug/sprint_gb3_b6_rung_memo.md`.*

5.1 The Stefan–Boltzmann constant lives in the temporal projection

Compactifying the temporal direction to S^1_β promotes the Casimir computation to the finite-temperature free energy. The high-temperature expansion contains the Stefan–Boltzmann term

$$\frac{F(\beta)}{V_3} \sim -\frac{\pi^2/90}{\beta^4} + \cdots, \qquad \beta \to 0, \tag{9}$$

where $V_3 = 2\pi^2$ is the volume of the unit S^3. The $\pi^2/90$ traces directly to the Matsubara sum: after analytic continuation, the temporal integral collapses to $\zeta_R(4) = \pi^4/90$, which is exactly the calibration tier of Paper 18 (even-zeta content from a second-order operator on a compact direction). The Matsubara prefactor $(2\pi/\beta)^{2s}$ from (5) provides one factor of π^{2s}, and the $\zeta_R(4)$ provides the rest; the net π-power survives to give the Stefan–Boltzmann constant.

Remark 2 (The high-T $\pi^2/90$ is Matsubara-induced)**.** *The $\pi^2/90$ in the Stefan–Boltzmann limit of the $S^3 \times S^1_\beta$ free energy is not a property of the spatial spectrum on S^3. It is the result of integrating the spatial spectrum against a Matsubara temporal sum, where the latter's 2π injection (Observation 2) propagates through the $\zeta_R(4)$-controlled spectral integral. Without the temporal compactification, the same spatial spectrum gives Casimir $1/240$ (Observation 3), with no π.*

Observation 5 (Two-window discriminator: each Bernoulli rung is lit from both sides by distinct physics (Sprint GB-4))**.** *Remark 2 is the B_4 instance of a general structure. The conformal-scalar Casimir on S^{2k-1} supplies the* skeleton *window $\zeta_R(1-2k)$ (vacuum, π-free); the Stefan–Boltzmann free energy on $S^{2k-1} \times S^1_\beta$ supplies the* transcendental *(M_2) window $\zeta_R(2k) = \text{rational} \cdot \pi^{2k}$ (thermal). The Bose–Einstein integral $\int_0^\infty x^d/(e^x-1)\,dx = \Gamma(d+1)\zeta_R(d+1)$ makes this explicit: $d = 3$ gives $\zeta_R(4) = \pi^4/90$ (B_4), $d = 5$ gives $\zeta_R(6) = \pi^6/945$ (B_6).*

rung	*skeleton (vacuum)* $\zeta(1-2k)$	M_2 $\zeta(2k)$ *(thermal)*
B_4	$\zeta(-3) = 1/120$, S^3 *Casimir*	$\zeta(4) = \pi^4/90$, $S^3 \times S^1$ *S–B*
B_6	$\zeta(-5) = -1/252$, S^5 *Casimir*	$\zeta(6) = \pi^6/945$, $S^5 \times S^1$ *S–B*

Each rung is thus lit from both *functional-equation windows by* physically distinct *observables — vacuum Casimir and thermal radiation — related by the temperature-inversion duality $T \leftrightarrow 1/T$, which is the ζ functional equation $\zeta(s) \leftrightarrow \zeta(1-s)$ on the thermal circle.*

Negative control (the discriminating half). *The genuinely combinatorial GeoVac invariants — those forced by the packing axiom rather than by sphere field theory — are* off *the ladder: $B = 42$,*

$\Delta = 1/40$ *(Paper 2), and* $\kappa = -1/16$ *are not* ζ*-rung values (and the correlation entropy* $S_{\rm full}(\mathrm{GS})$ *is PSLQ-null off the master Mellin engine, Sprint TD Track 5). So the ladder is the functional-equation structure of the framework's* spectral-geometry *sector (Casimir + thermal + heat-kernel),* not *a universal property of every GeoVac quantity.*

Honest scope. *Both windows are lit by sphere field theory, so the positive landings are textbook in isolation (Planck blackbody, Casimir, Cardy temperature inversion); the GeoVac content is (i) that the framework's gravity and thermal sectors realize this two-window structure at three rungs, and (ii) the clean boundary against the combinatorial core. This sharpens, but does not exceed, the structural correspondence of Paper 32's Bernoulli-ladder remark; the non-trivial zeros remain untouched (Paper 28's functional-equation obstruction). As a cross-connection, the on-ladder/off-ladder split (*$F = \zeta(2)$ *on,* B *and* Δ *off) is exactly the Phase-4G "no common generator" decomposition of* $K = \pi(B + F - \Delta)$ *viewed from the ladder. Source:* `debug/sprint_gb4_discriminator_memo.md`.

6 Implications for Paper 34

Paper 34's projection dictionary, in the form current when this sprint ran, did not separately index "introduce a mass parameter into the spectrum" from "compactify the temporal direction for finite-time observation." In practice these had been bundled with Fock-conformal (which carries Z and E) or with the spectral action (which carries Λ). This sprint shows the bundling is too coarse: the mass projection is ring-preserving, while the observation/temporal-window projection injects 2π *at the spectrum level, not just at the integration level.* These are structurally different and should be split.

Proposition 1 (Two new rows for the Paper 34 projection dictionary)**.** *The Paper 34 projection list should include the following two explicit rows:*

- ***Rest-mass projection*** *(§ XIV.A). Source: bare KG spectrum on* $S^3 \times \mathbb{R}$*. Target: massive KG spectrum* $\omega_n^2 = n(n+2)/R^2 + m^2$*. Variables introduced:* m *(mass; equivalently rest-energy* mc^2 *at* $c = 1$*). Dimension introduced: mass = energy at* $c = 1$*. Transcendental signature:* trivial*; the bare-graph algebraic-extension ring is preserved when* $m^2 \in \mathbb{Q}$ *(or any ring extension of* $\mathbb{Q}$ *already present in the spatial spectrum). Used in: muonic / electronic / positronium reduced-mass corrections; nuclear-electronic composed Hamiltonian (Paper 23 hyperfine I.S coupling); any future Klein-Gordon or Dirac-on-*S^3 *relativistic mass calculation.*
- ***Observation / temporal-window projection*** *(§ XIV.B). Source: KG spectrum on* $S^3 \times \mathbb{R}$*. Target: KG spectrum on* $S^3 \times S^1_\beta$ *with periodic Euclidean time of period* β *(equivalently, a finite-time observation window of duration* β *in real time). Variables introduced:* β *(or temperature* $T = 1/\beta$*). Dimension introduced: time (= inverse energy). Transcendental signature:* $2\pi \cdot \mathbb{Q}$ *per Matsubara mode in the spectrum directly; in spectral integrals over the compactified direction,* $\pi^{2k} \cdot \mathbb{Q}$ *from* $\zeta_R(2k)$ *values, including the Stefan-Boltzmann* $\pi^2/90$*. Used in: all finite-temperature observables (Casimir at* $T \neq 0$*, free-energy density, Stefan-Boltzmann constants); all QED loop integrals that involve continuous time integration; any spectral-action evaluation of* $\mathrm{Tr}\, f(D^2/\Lambda^2)$ *where the trace is over a compact-time direction.*

The two rows together replace the implicit "mass / observation" content that has been bundled with other rows. They make explicit a distinction that is structural in the framework but had not been named. The Casimir computation in Section 5 is the cleanest empirical demonstration: changing the temporal boundary condition (open $\mathbb{R} \to$ closed S^1_β) changes the transcendental class of the computed observable from rational (1/240) to $\pi^{2k}\cdot\mathbb{Q}$ (Stefan-Boltzmann), without any change to the spatial spectrum.

Remark 3 (This is a no-edits-yet proposal)**.** *Per the Paper 34 maintenance protocol, the addition of two new rows to the projection dictionary is a substantive structural change and requires PI direction. The PM may append match rows to Paper 34's empirical catalogue autonomously, but new projection-list rows are not PM-autonomous. The present paper records the structural finding and the proposal; Paper 34 itself is not amended.*

7 Falsifiable prediction

Prediction 1 (π enters at temporal-projection compactifications, only)**.** *A GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter that has been promoted from the discrete graph spectrum. Equivalently: the discrete graph itself is π-free; the projection that* integrates *(over time, over a Matsubara mode, over a Mellin parameter, over an analytically continued spectral mode) is where π appears.*

The closest theorem-grade analog in the continuum literature is the **Bisognano–Wichmann theorem** [21] (see [19] for a contemporary geometric account in the Morinelli–Neeb–Ólafsson lineage and [20] for the non-unitary CFT extension): the BW theorem identifies the modular automorphism of a wedge-restricted field algebra with the Lorentz-boost flow at rapidity 2π per unit proper time, which is exactly the compactification operation Prediction 1 flags as the unique π-source. Section 8.6 below develops this identification explicitly, and Paper 32 § VIII makes the spectral-triple-side case-exhaustion statement that subsumes the prediction structurally. The reader who treats Prediction 1 as a discrete-substrate shadow of BW will recover the falsifier (§ 8.6 Track D #1) and the 208/208 verification panel below from a single conceptual root.

7.1 Confirmatory evidence

Six existing GeoVac observables that do not require integration over a continuous parameter are π-free, as predicted (the sixth was added after sprint KG-5):

1. **Pauli term counts** (Paper 14): the universal $N_{\text{Pauli}} = 11.10 \cdot Q$ rule across 35 composed molecules is integer/rational at every Q.
2. **Nuclear magic numbers** (Paper 23): $\{2, 8, 20, 28, 50, 82, 126\}$ from HO + spin-orbit graph eigenvalue counting; all integers.
3. **Hydrogenic atomic eigenvalues via the Fock/S^3 projection** (Paper 7): $\lambda_n = -(n^2-1)$ for the Fock-projected Coulomb operator (a continuum S^3 Laplace–Beltrami property, not a bare-graph one); pure integers at every n.
4. **Bargmann–Segal HO graph** (Paper 24): bit-exactly π-free in exact rational arithmetic at every finite $N_{\max}$, verified for $N_{\max} \leq 5$.
5. **Graph-native QED matrix elements** (Papers 28, 33): every matrix element in $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$ at every finite $n_{\max}$.
6. **Spatial Dirac Casimir on unit** S^3 (Section 9 item 1): $E_{\text{Cas}}^{\text{Dirac}} = 17/480$, exact rational, computed in sprint KG-5 against the Camporesi–Higuchi spectrum. Confirms the prediction in the spinor sector.

7.2 Counter-examples (also confirmatory)

Three existing GeoVac observables that *do* contain π, all of which arise from a temporal or spectral continuous integration:

1. **Vacuum polarization** $\Pi = 1/(48\pi^2)$ (Paper 28): one-loop QED on S^3, evaluated as a continuous spectral integral over the Schwinger proper time $t \in (0, \infty)$. The proper-time integration is the temporal-projection step; the π^2 traces to the Seeley-DeWitt coefficient a_2 on $S^3 = \sqrt{\pi}/8$, squared.
2. **QED β-function** $2\alpha^2/(3\pi)$ (Paper 28): same provenance. The π in the denominator is the same $\sqrt{\pi}$ factor from the Seeley-DeWitt coefficients.
3. **Stefan-Boltzmann** $\pi^2/90$ (Section 5): high-T limit of the $S^3 \times S^1_\beta$ free energy density; π from the Matsubara $\zeta_R(4)$ sum.

In every case, the π-bearing observable can be traced to a specific continuous integration, and the integration can be identified with the observation/temporal-window projection of Proposition 1. The framework so far has no exception to the prediction.

7.3 Sprint TX-B verification panel (May 2026)

Sprint TX-B (May 2026) tested Prediction 1 on five new observables, with predictions logged a priori in `debug/data/tx_b_predictions.json` *before any computation script was run.* This discipline guards against retroactive reframing, the standard failure mode for falsifiable predictions (cf. `docs/curve_fit_audit_memo.md`, 2026-05-02). The panel was designed to probe four corners of the prediction:

1. **Stefan-Boltzmann coefficient on** $S^3 \times S^1_\beta$ (positive control, π predicted): re-derives $\pi^2/90$ from the Matsubara sum. Confirmed.
2. **Spatial Casimir on Bargmann–Segal S^5 for the 3D HO** (positive prediction, no π): the Bargmann construction is a different geometry from the S^3 Coulomb / S^3 Dirac sectors of KG-3 / KG-5. If the half-integer Hurwitz mechanism is structural, the S^5 HO Casimir should produce another exact rational. Computed via the spectral zeta

$$\zeta_X(s) = \tfrac{1}{2}\left[\zeta_R(s{-}2,\, 3/2) - \tfrac{1}{4}\zeta_R(s,\, 3/2)\right]$$

at $s = -1$. Bernoulli-at-half-integer values $B_2(3/2) = 11/12$, $B_4(3/2) = 127/240$ give Hurwitz at negative integers $\zeta_R(-1, 3/2) = -11/24$, $\zeta_R(-3, 3/2) = -127/960$, and the bracket $[\zeta_R(-3, 3/2) - \frac{1}{4}\zeta_R(-1, 3/2)] = -17/960$. Hence

$$E_{\text{Cas}}^{\text{HO},S^5} \;=\; \tfrac{1}{2}\,\zeta_X(-1) \;=\; -\tfrac{17}{3840} \quad \text{(exact rational, no } \pi\text{)}.$$

This is a clean new closed form, structurally a sibling of KG-3 (scalar S^3 Casimir = 1/240) and KG-5 (Dirac S^3 Casimir = 17/480). The shared mechanism across all three sectors — half-integer Hurwitz at shift $a = 3/2$, Bernoulli polynomials at 3/2 rational — is the universal structural driver: any spectrum of the form $E_n = n + a$ with $a \in \mathbb{Q}$ and degeneracy polynomial in n produces a rational Casimir through ζ-regularization, regardless of whether the underlying manifold is S^3 scalar, S^3 Dirac, or S^5 HO. The Coulomb/HO/Dirac asymmetry that Paper 24 highlights at the level of the Laplacian role *does not* extend to the Casimir transcendental class. Confirmed.

3. **Hydrogen Bohr spectrum** (negative control): pure algebraic formula, no integration, no π. Trivial but essential. Confirmed.
4. **Heisenberg-Euler 1-loop coefficient** (positive prediction): the textbook leading coefficient $\alpha^2/(45\pi m_e^4)$ contains π in the denominator, sourced by the Schwinger proper-time integration $\int_0^\infty ds/s^3\, e^{-m^2 s} f(eF, s)$ — the canonical example of an observation/temporal-window projection. We did not derive the absolute coefficient natively from the GeoVac framework (that would require LS-7-style two-loop SE machinery), so this entry is flagged as structural identification (error code S in Paper 34's classification): the transcendental class matches, the absolute value is from the textbook. Confirmed at structural level.
5. **Static Coulomb-like potential on the Hopf S^3 graph** (positive prediction, no π): the regularized discrete Green's function $(L+I)^{-1}$ at finite $n_{\max}$ is built by exact matrix inversion on a finite integer-valued graph. No continuous integration appears. Computed in exact sympy rational arithmetic at $n_{\max} = 2, 3$: every entry is a sympy `Rational`, no π anywhere. The continuum limit ($n_{\max} \to \infty$) would inject π via spherical-harmonic normalization on S^3, exactly where Prediction 1 predicts. Confirmed at finite truncation.

Tally: 5 of 5 confirmed. Combined with the prior KG sprint (200 KG spectrum cases, KG-2 temporal injection, KG-3 scalar Casimir, KG-5 Dirac Casimir), Prediction 1 has now been tested on a cumulative **208 individual checks across the KG and TX-B sprints, with 100% match rate**.

Remark 4 (graduation to load-bearing principle)**.** *With the TX-B verification panel in hand, Prediction 1 graduates from a falsifiable* conjecture *of the framework to a* load-bearing principle *of it. The graduation is two-fold: (i) the prediction has survived 208 independent checks across two sprints, with no exceptions; (ii) it now plays an operational role alongside Paper 18's transcendental cataloguing as a discipline for evaluating new derivations. If a step in a derivation is supposed to produce π, the discipline is to identify the continuous integration over a temporal or spectral parameter that produces it; if no such integration is identifiable, the π is either an error or an unidentified projection (in the latter case requiring an addition to Paper 34's projection list).*

7.4 Falsification criterion

A counter-example to Prediction 1 would be a GeoVac observable whose evaluation does not include a continuous integration over a temporal or spectral parameter, but which contains π. The most natural place to look is the Paper 2 Observation $K = \pi(B + F - \Delta) \approx 1/\alpha$, in which the π is explicit on the right-hand side and the projections involved (finite Casimir trace at $m = 3$, infinite scalar Dirichlet at $s = d_{\max}$, single-level Dirac mode degeneracy) all appear, on present analysis, to be discrete-graph quantities. If the projection-depth analysis of Paper 34 § 9 item 3 is correct, then either (a) the π in K is not what it looks like (i.e., it factors out cleanly as a temporal-integration prefactor), or (b) the present prediction is false and there is a non-temporal route to π in the framework that we have not identified. Either resolution would be informative.

7.5 Promotion to a spectral-triple theorem (Sprint TS, May 2026)

Track TS-D of Sprint TS (May 2026; memo `debug/track_ts_d_paper35_triple_test_memo.md`) tested Prediction 1 against each of Paper 34's fifteen then-catalogued projections through the spectral-triple lens of Paper 32 [10], and found that every π-source identifiable in any projection reduces to one of three triple-framing mechanisms: M1 (Hopf-base measure $\mathrm{Vol}(S^2)/4$), M2 (Seeley–DeWitt heat-kernel coefficient $\sqrt{\pi}$ on S^3), or M3 (vertex-topology Hurwitz Dirichlet L-content).

Track TS-E1 sharpened the case exhaustion to a formal theorem statement, now stated and proven in Paper 32 § VIII as the *π-source case-exhaustion theorem*. Track TS-E3 further showed that M1, M2, and M3 are not three independent mechanisms but three sub-cases of a single master Mellin mechanism $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ parametrized by operator order $k \in \{0, 1, 2\}$ — the master-mechanism reading is recorded in Paper 18 § III.7.

The case-exhaustion theorem is proven *by exhaustion over the Paper 34 projections § III.1–§ III.15* (the fifteen then catalogued), for which "the converged value contains π", "the chain contains a continuous spectral/temporal integration", and "the chain engages one of M1/M2/M3" are shown equivalent. *Within that scope*, Prediction 1 is therefore a structural corollary of the GeoVac spectral triple's three operating modes (Hopf decomposition of the spinor bundle, heat kernel of D^2, vertex-parity selection on the Camporesi–Higuchi spectrum), rather than a bare empirical regularity. For observables outside the theorem's enumerated scope (the projections added since, and the general biconditional over *all* GeoVac observables) the reading remains the *structural Observation* of this paper — WH7 is registered, not proven — with the 208/208 confirmation as its empirical foundation; the theorem supplies the structural "why" where it applies.

8 Tensor-product verification on $T_{S^3} \otimes T_{S^1_\beta}$

Sprint TD Track 1 (May 2026) constructs the explicit tensor-product spectral triple $T_{S^3} \otimes T_{S^1_\beta}$ with Dirac operator

$$D_{\text{total}} \;=\; D_{S^3} \otimes 1 \;+\; \gamma_{S^3} \otimes D_\tau, \tag{10}$$

where D_τ on S^1_β is the Matsubara operator with bosonic spectrum $\omega_k = 2\pi k/\beta$ or fermionic spectrum $\omega_k = 2\pi(k + 1/2)/\beta$. The anti-commutator $\{D_{S^3} \otimes 1, \gamma_{S^3} \otimes D_\tau\} = 0$ makes D^2_{total} block-diagonal (no cross term), so spectra add and the heat kernel factorises as $\mathrm{Tr}\, e^{-sD^2_{\text{total}}} = K_{S^3}(s) \cdot K_{S^1_\beta}(s)$.

8.1 Stefan–Boltzmann as exact $M_1 \times M_2$

The textbook free-energy density for a single bosonic massless scalar on flat $\mathbb{R}^3 \times S^1_\beta$ is $F/V = -(\pi^2/90)\, T^4$. The high-T continuum limit on $S^3 \times S^1_\beta$ recovers the same coefficient. Sprint TD Track 1 derives this as a single SYMPY-exact identity:

$$\frac{F}{V} \;=\; -\underbrace{\frac{1}{2\pi^2}}_{M_1\ \text{Hopf measure}} \cdot \frac{1}{3} \cdot \underbrace{6\,\zeta(4)}_{M_2\ \text{Riemann}\ \zeta} \cdot T^4 \;=\; -\frac{\pi^2}{90}\, T^4 \tag{11}$$

with sympy residual 0. The $1/(2\pi^2)$ is the M_1 Hopf-base measure factor $\mathrm{Vol}(S^2)/(2\pi)^3$; the $6\zeta(4) = \pi^4/15$ is the M_2 Riemann $\zeta(4)$ at the Bose–Einstein integrand. The net π^2 in $-\pi^2/90$ is the $M_1 \times M_2$ product in π-power arithmetic ($\pi^{-2} \cdot \pi^4 = \pi^2$). For Dirac fermions, the same factorisation with the fermionic $\eta(4)/\zeta(4) = 7/8$ ratio gives $F/V_{\text{per-Weyl}} = -(7\pi^2/720)\, T^4$, again with residual 0.

8.2 Spatial Casimir is rational on both bundles

In the zero-T limit the thermal partition function reduces to the spatial Casimir on S^3, recovering Paper 35 KG-3 and KG-5 verbatim:

$$E_{\text{Cas}}^{\text{conformal scalar}, S^3} = \tfrac{1}{240},$$
$$E_{\text{Cas}}^{\text{full Dirac}, S^3} = \tfrac{17}{480},$$

both *rational*. The half-integer Hurwitz / Bernoulli-polynomial collapse mechanism noted in §V transports verbatim to the tensor product: the spatial factor contributes no π at the discrete-mode level, and π enters only through the Mellin transform on s when the heat kernel is evaluated.

8.3 Modular residual on the tensor product

The Sprint MR-B closed-form modular residual on the spatial Dirac sector,

$$\varepsilon(t) \;=\; \sum_{m\geq 1}(-1)^m \sqrt{\pi}\, e^{-m^2\pi^2/t}\left[t^{-3/2} - 2m^2\pi^2\, t^{-5/2} - \tfrac{1}{2}\, t^{-1/2}\right], \tag{12}$$

transports verbatim to the tensor product (it lives on K_{S^3}). The temporal factor contributes a leading $K_{S^1_\beta}(s) \approx \beta/(2\sqrt{\pi s})$ from Jacobi-theta inversion. Crucially, the $\sqrt{\pi}$ in $K_{S^1_\beta}$ is the M_1 $\mathrm{Vol}(S^1)$ / *Mellin-measure* factor, whereas the $\sqrt{\pi}$ in $\varepsilon(t)$ is the M_2 *Seeley–DeWitt* prefactor on S^3. Same numerical constant; structurally distinct mechanisms; distinct tensor factors. The ring of the full modular residual is $\sqrt{\pi}\cdot\mathbb{Q}\oplus\pi^2\cdot\mathbb{Q}$ (all M_2) on the spatial side, multiplied by $\sqrt{\pi}/\beta\cdot\mathbb{Q}$ (M_1) on the temporal side.

8.4 Transcendental audit holds Prediction 1

Sprint TD Track 1 audits ten distinct transcendental sources in the construction. By tensor factor: 4 π-bearing on S^3 (Hopf measure, $\mathrm{Vol}(S^3)/\pi^2$, M_2 modular exponent π^2, M_2 Seeley–DeWitt prefactor $\sqrt{\pi}$); 5 on S^1_β (Matsubara modes, Bose–Einstein integrand, theta-inversion prefactor, Hopf-measure factor, fermionic $\eta(4)$ ratio); 3 rational with no π (scalar Casimir 1/240, Dirac Casimir 17/480, fermionic 7/8 ratio). By mechanism: 3 M_1, 4 M_2, 0 M_3 (no vertex coupling in the free thermal partition function), 1 $M_1\times M_2$ product (Stefan–Boltzmann), 2 rational collapses (M_2 at half-integer Hurwitz).

Every π has a continuous-integration source: Matsubara compactification of imaginary time, Weyl-asymptotic spatial integration, Mellin transform of the heat-kernel trace, Jacobi-theta inversion, anti-periodic boundary condition. Every rational result comes from a discrete spectral evaluation: Bernoulli polynomials at $a = 3/2$, Dirichlet $\eta(4)/\zeta(4)$ rational ratio, half-integer Hurwitz collapse. **Prediction 1 holds verbatim on the tensor product.**

8.5 Honest scope

This subsection is a verification of the master Mellin engine partition on a known thermal field-theory example. The Stefan–Boltzmann constant is textbook; the new content is the explicit factor-by-factor structure that pins each π to its tensor factor and to its Mellin mechanism. The construction is at the spectrum level; a full spectral-triple-of-tensor-products theorem in the sense of Paper 39 (which handles the same-manifold case $T^{\lambda_a}_{S^3}\otimes T^{\lambda_b}_{S^3}$) is not claimed for the cross-manifold $S^3\times S^1_\beta$ case. See Paper 32 §VIII.D on the cross-manifold open question. Sprint TD Track 1's `geovac/thermal_tensor_triple.py` (SYMPY symbolic, 36 tests) is the computational anchor.

8.6 Bisognano–Wichmann reading of the temporal-window projection

Sprint TD Track 4 (`debug/sprint_td_track4_memo.md`, May 2026) extends the spectrum-level construction of § 8 to the Euclidean Schwarzschild cigar. Smoothness at the horizon $r = 2M$ (the standard Hawking–Gibbons absence-of-conical-defect argument) forces the imaginary-time circle to have period $\beta_{\mathrm{cigar}} = 8\pi M$, reproducing the Hawking temperature $T_H = 1/(8\pi M)$ from the M1

Hopf-base measure / Vol(S^1_τ) mechanism alone. Sprint TD Track 4's `matsubara_spectrum()` called with $\beta = 8\pi M$ reproduces the standard Hawking spectrum bit-identically (lowest bosonic mode $1/(4M) = \kappa_g$ surface gravity; lowest fermionic πT_H); sympy residual zero on all three identities ($\beta = 8\pi M$, $T_H = \kappa_g/(2\pi)$, $\beta = 4M \cdot 2\pi$).

Track D (`debug/bisognano_wichmann_track_d_memo.md`, May 2026) makes explicit the Bisognano–Wichmann reading of this result. The Bisognano–Wichmann theorem [21] (see also [19] for a contemporary geometric perspective in the Morinelli–Neeb–Ólafsson lineage) states that the modular automorphism of the algebra of observables localized in a Rindler wedge, with respect to the vacuum state, IS the unitary representation of the Lorentz boost subgroup that maps the wedge to itself; the modular Hamiltonian K generates the boost, and the Tomita–Takesaki KMS condition forces the analytic period $\sigma_{i\cdot 2\pi}$ = identity. Sewell 1982 [22] generalized this to globally hyperbolic Lorentzian manifolds with bifurcate Killing horizons: the natural Hartle–Hawking vacuum [23] restricted to the Schwarzschild exterior is a KMS state at inverse temperature $\beta = 8\pi M$, and the Tomita modular automorphism is the surface-gravity-rescaled Killing flow. Hartle–Hawking [23] provided the path-integral derivation of the same imaginary-time period $8\pi M$.

The structural identification. Under the Wick-rotation chain [Hartle–Hawking 1976] $\to$ [Sewell 1982] $\to$ [Bisognano–Wichmann 1976], the framework's M1 2π on S^1_τ IS the period of the modular-flow / Lorentz-boost orbit of a stationary observer outside the Schwarzschild horizon. Equivalently, the temporal-window projection (Paper 34 §III.15), instantiated at $\beta = 8\pi M$, has dual physical interpretations: a Hopf-bundle measure factor on the temporal S^1_τ (Riemannian) and a modular-flow / Lorentz-boost orbit period (Lorentzian, via the standard Wick rotation). This is a *structural correspondence*, not a literal identification: two mathematical objects (the framework-side Hopf-base measure factor; the Lorentzian-side modular-automorphism period) are equated by the published Wick-rotation prescription, not by an operator-level proof inside the spectral triple. The identification stands on the published QFT machinery, which is mature and verified independently of GeoVac.

Unruh corollary. The same M1 mechanism applied to a Rindler wedge gives the Unruh effect: a uniformly accelerated observer at proper acceleration a has Wick-rotated $\beta_{\text{Rindler}} = 2\pi/a$, reproducing $T_U = a/(2\pi)$ from M1. Sprint TD Track 4's `matsubara_spectrum()` machinery applies verbatim with β_{Rindler} in place of β_{cigar}; explicit verification is a sprint-scale pendant follow-up flagged in Track D §5.4.

Honest scope. Track D commits to four claims: (1) the framework's M1 mechanism produces $\beta = 8\pi M$ on the cigar (sympy residual zero, Sprint TD Track 4); (2) this β is the imaginary-time period of the Hartle–Hawking thermal state; (3) Sewell 1982 proves this thermal state IS a Bisognano–Wichmann modular flow; (4) the framework-side 2π and the Lorentzian-side modular-flow 2π are the same number, by the published Wick-rotation chain. It does *not* claim: an operator-level proof that the modular Hamiltonian on the truncated metric spectral triple closes at period 2π (this is the principal falsifier, a follow-up beyond sprint scale); a Lorentzian-signature spectral triple (a major NCG structural extension, Track B verdict NO-GO; `debug/lorentz_boost_scoping_memo.md`); a derivation of the cigar geometry from the GeoVac packing axiom (Sprint TD Track 4 V3 NEGATIVE; the cigar is forced by Einstein equations + Schwarzschild horizon, not by GeoVac); Bekenstein–Hawking entropy from the horizon S^2 (out of scope, Sprint TD Track 4 V2 deferred).

Metric-level dual (Paper 45 § open Q1, 2026-06). The temporal-compactification picture here has a metric-level counterpart: the truncated Bisognano–Wichmann boost is *compact* (integer spectrum, $e^{2\pi i K} = I$ at every finite cutoff, the same period-2π modular closure), so the finite-cutoff quantum metric on the truncated Krein triple is Euclidean (forward-triangle), and the Lorentzian signature is the Wick rotation of the compact 2π-circle — a convention, with the genuine reverse-triangle signature confined to the non-compact continuum limit. Just as compactification is the one step that injects π into the *spectrum* (this paper), it is the one step that fixes the *signature* as a Wick-rotation convention: the same compactness $\Rightarrow$ discreteness $\Rightarrow$ π/β reading on both faces.

Cross-references. Paper 32 §VIII Remark `rem:bisognano_wichmann_reading` for the spectral-triple-side statement and case-exhaustion-theorem context; Paper 34 §III.15 footnote for the temporal-window projection's Bisognano–Wichmann reading; Paper 34 §V.B for the Hawking-T off-precision catalogue row (error class C, M external Einstein-equations input); Paper 45 § open Q1 for the metric-level dual above.

8.7 Unruh pendant: the third Wick-rotation face

The same M1 mechanism applied to a Rindler wedge gives the Unruh effect, the flat-space analog of Hawking radiation and the third face of the Wick-rotation chain that § 8.6 traversed in the Schwarzschild geometry. Sprint Unruh-Pendant (`debug/unruh_pendant_memo.md`, May 2026, a sprint-scale pendant follow-up to Track D) documents the construction.

Wick rotation of Rindler. A uniformly accelerated observer at proper acceleration a traces a hyperbolic worldline in the right Rindler wedge $x > |t|$ of Minkowski space; in Rindler coordinates (η, ρ) defined by $t = \rho \sinh(a\eta)$, $x = \rho \cosh(a\eta)$, the wedge metric is $\mathrm{d}s^2 = -\rho^2\,\mathrm{d}\eta^2 + \mathrm{d}\rho^2 + \mathrm{d}y^2 + \mathrm{d}z^2$. Wick-rotating the rapidity $\eta \to i\eta_E$ produces the Euclidean polar metric $\mathrm{d}s_E^2 = \rho^2\,\mathrm{d}\eta_E^2 + \mathrm{d}\rho^2 +$ (transverse). Smoothness at $\rho = 0$ (the standard absence-of-conical-defect argument, identical in form to the Hawking–Gibbons argument for the Schwarzschild cigar tip) forces η_E to have period 2π. The accelerated observer's proper time $\tau_p = \eta_E/a$ at fixed $\rho = 1/a$ therefore has period

$$\beta_U = \frac{2\pi}{a}, \qquad T_U = \frac{a}{2\pi}, \qquad \text{equivalently} \quad T_U = \frac{\hbar a}{2\pi c k_B} \text{ (SI units).}$$

M1 identification. The single 2π in β_U is exactly the Hopf-base measure $\mathrm{Vol}(S^1) = 2\pi$ on the angular η_E-circle, identical in structural role to the 2π in Sprint TD Track 4's cigar τ-period and in Track 1's Matsubara S^1_β-circumference. Sprint TD Track 1's `matsubara_spectrum()` called with $\beta = 2\pi/a$ reproduces the Unruh-thermal Matsubara spectrum bit-identically (manual sympy check, residual zero): lowest non-zero bosonic mode equals a (the Unruh analog of Hawking's κ_g surface gravity); lowest fermionic mode equals $a/2 = \pi T_U$; T_U from $1/\beta_U$ identifies bit-exactly with $a/(2\pi)$.

The unified Wick-rotation theorem. The four-witness chain

> *Hartle–Hawking 1976 path-integral $\to$ Sewell 1982 generalization to bifurcate Killing horizons $\to$ Bisognano–Wichmann 1976 Rindler-wedge modular flow $\to$ Unruh 1976 flat-space limit*

is one Wick-rotation theorem with four physical instantiations: the thermal state of an observer with restricted causal access is determined by the regularity period of the Euclidean rotation of their causal-domain coordinates, which equals 2π times 1/(surface gravity). Surface gravity values in the four witnesses: $\kappa_g = 1/(4M)$ (Schwarzschild horizon), a (Rindler / Unruh), the boost-rapidity rate (Bisognano–Wichmann), and the Killing surface gravity in the Sewell-generalized statement. In each case, the 2π is the $\mathrm{Vol}(S^1)$ Hopf-base measure of the master Mellin engine at $k = 0$ (M1).

Honest scope. Same as § 8.6. The framework's M1 mechanism on $S^1_{\eta_E}$ IS the Wick-rotated image of the modular flow on the Rindler wedge, via the published Wick-rotation prescription (Unruh [24]; Bisognano–Wichmann [21]; Sewell [22]; Hartle–Hawking [23]). This is a *structural correspondence*, not a literal identification: the Hopf-bundle measure factor on the Riemannian side and the modular-automorphism period on the Lorentzian side are equated by the published Wick-rotation chain, not by an operator-level proof inside the spectral triple. The natural Unruh-side falsifier is structurally the same as Track D's falsifier #1, instantiated at the Rindler wedge instead of the Schwarzschild static exterior: compute the modular Hamiltonian on the truncated metric spectral triple $\mathcal{T}_{n_{\max}}$ for a Rindler-wedge-restricted Camporesi–Higuchi state and verify $\sigma_{i\cdot 2\pi} =$ identity at finite $n_{\max}$. *One falsifier covers both faces* (and also Hawking): the chain that ties the framework 2π to the modular-flow 2π runs through the same KMS-Tomita-Takesaki algebra in every case, so lifting the period-closure check on any one face lifts the structural correspondence on all three. A follow-up beyond sprint scale; priority MEDIUM-HIGH.

Cross-references. Paper 32 §VIII Remark `rem:bisognano_wichmann_reading` (Unruh paragraph appended) for the spectral-triple-side statement; Paper 34 §III.15 footnote (extended with Unruh closure of the Track D §5.4 follow-up flag); Paper 34 §V.B Unruh-T off-precision catalogue row (machinery-witness, error class C, a external proper-acceleration input); Paper 34 §VIII Lorentz-boost open question (Track D documentation closure paragraph extended with Unruh four-witness completion).

9 Open questions

1. **Spinor sector (resolved May 2026, sprint KG-5).** Computed. The Dirac-on-S^3 spectrum $|\lambda_n| = n + 3/2$ with degeneracy $g_n = 2(n+1)(n+2)$ (Camporesi–Higuchi [18]) gives the spectral zeta
$$\zeta_{|D|}(s) \;=\; 4\big[\zeta_R(s{-}2,\, 3/2) \;-\; \tfrac{1}{4}\zeta_R(s,\, 3/2)\big]$$
in the full-Dirac convention (both chiralities, both eigenvalue signs, complex spinor bundle of dimension 4 on S^3). Hurwitz values via Bernoulli polynomials at $a = 3/2$: $B_2(3/2) = 11/12$, $B_4(3/2) = 127/240$. Substitution gives $\zeta_{|D|}(-1) = -17/240$, hence
$$\boxed{E_{\text{Cas}}^{\text{Dirac}} \;=\; -\tfrac{1}{2}\zeta_{|D|}(-1) \;=\; \tfrac{17}{480}} \tag{13}$$
Exact rational, no π, matches the textbook value of Camporesi–Higuchi 1996 (Eq. 5.27), Bytsenko et al. [15] § 4.5, and Dowker–Altaie 1978 (Phys. Rev. D **17**, 417), the vacuum-averaged stress-energy tensor for spinor fields in the Einstein universe, which gives $\langle T_{00}\rangle = 17/(1920\pi^2 a^4)$ (a value that paper attributes to Ford); integrated over $\mathrm{Vol}(S^3) = 2\pi^2 a^3$ this is $E = 17/(960a)$, matching the 17/960 single-Weyl-chirality convention line quoted below — the density numeral 1920 and the energy numeral 960 differ by the volume factor and should not be identified. (Its finite-temperature companion, Altaie–Dowker, Phys. Rev. D **18**, 3557, was cited here between 2026-08-28 and this correction; the zero-temperature member of the pair is the right one for a Casimir energy, and the author order differs between them.) Verified to 40 dps numerically. (Attribution corrected 2026-08-28: this sentence previously cited Kennedy–Critchley–Dowker 1980, which is Ann. Phys. **125**, 346 — finite-temperature *scalar* field theory *with boundaries*, neither spinorial nor on the boundaryless S^3. The section number was also inconsistent with this paper's own bibliography entry.) Convention translations (single Weyl chirality with both signs: 17/960; single Majorana real d.o.f.: 17/960; single chirality, single sign: 17/1920) are catalogued in `debug/data/kg5_dirac_casimir.json`.

   The per-step transcendental ledger is identical in structure to the scalar case (Section 5): zero π injections in the spatial calculation; the half-integer Hurwitz shift stays in $\mathbb{Q}$ because Bernoulli polynomials have rational coefficients and $3/2 \in \mathbb{Q}$. The Paper 35 prediction is **confirmed in the spinor sector**.

   For the temperature-corrected case on $S^3 \times S^1_\beta$, fermions require antiperiodic (Neveu-Schwarz-like) boundary conditions on the temporal circle; the Matsubara modes shift to $\omega_k^t = (2k +$

$1)\pi/\beta$ instead of $2\pi k/\beta$. π enters at the lowest mode $\omega_0^t = \pi/\beta$ (no zero mode). The injection mechanism is structurally identical to KG-2 — the π is the radian/circumference ratio of the temporal S^1 — shifted by half a Matsubara unit because of the spin structure.

2. **Where does this leave the K combination rule? (Sprint K-CC, clean negative.)** The Paper 2 formula $K = \pi(B + F - \Delta) \approx 1/\alpha$ is the framework's most extreme multi-projection coincidence. Under Prediction 1, the explicit π on the right-hand side flagged two candidate readings: (i) K is, despite appearances, a single-projection statement of the spectral-action / temporal-compactification kind (cf. the Cardano root $\Lambda_\infty = 3.7102\ldots$ from the SD two-term exactness theorem; the π would then be the Hopf-base measure $\mathrm{Vol}(S^2)/4$, identified in Sprint A), or (ii) the prediction is false in the multi-loop / infinite-Dirichlet sector, and the π has a non-temporal source that the framework has not yet named. Sprint K-CC (May 2026) tested possibility (i) along three independent axes with a clean triple negative:

   (a) PSLQ at 100 dps against a 244-element basis of framework invariants (B, F, Δ, κ, $|\lambda_n|$, g_n, half-integer Hurwitz values, Bernoulli at half-integers, S^3 Seeley–DeWitt coefficients, π powers, and small-exponent products) returned **no identification of** Λ_∞. The depressed cubic $2\Lambda^3 - \Lambda - 4K/(\pi\sqrt{\pi}) = 0$ is recovered as positive control; removing K from the basis kills the identification entirely.

   (b) Search for a natural selection criterion for Λ from S^3 spectral data (35 cutoff–target pairs, 37 composite-eigenvalue candidates, plus critical-point analysis) returned only tautological "hits" (Gaussian trace at Λ_∞ literally equals K/π by construction). The cleanest non-tautological near-miss — the Gaussian trace equating to $B + F$ — gives Λ at relative distance 1.86×10^{-4} from Λ_∞, $\sim$three orders of magnitude looser than the K residual itself (4.8×10^{-7} for the plain sum; 8.8×10^{-8} via Paper 2's self-consistency root $1/\alpha + \alpha^2 = K$), and so cannot explain the K precision even if elevated.

   (c) The unified heat-kernel approach is blocked by **Theorem T9** (Paper 28): $\zeta_{D^2}(s)$ at integer s is constrained to $\sqrt{\pi}\cdot\mathbb{Q}\oplus\pi^2\cdot\mathbb{Q}$. Symbolic computation gives $\zeta_{D^2}(2) = \pi^2 - \pi^4/12$ and $\zeta_{D^2}(4) = \pi^6(168 - 17\pi^2)/1260$. $F = \pi^2/6$ does not appear at any integer s of the unified expansion. Independently of PSLQ, this is an *algebraic obstruction* to the unified-CC reading: B, F, Δ live in three categorically different spectral objects on two different bundles (finite scalar Casimir trace at $m = 3$; infinite scalar Fock–Dirichlet sum at $s = d_{\max}$; single-level Dirac mode degeneracy at $n = 3$ on the spinor bundle), and no single CC heat-kernel expansion contains all three as different terms.

   Reading (i) is therefore ruled out at the framework's current resolution. The remaining live readings are (ii) revised: the π in K has a non-temporal source the framework has not yet named, or (iii) K is a coincidence below the framework's intrinsic resolution and Prediction 1 is consistent but non-predictive at the 10^{-8} level. The K-CC sprint independently verified the closed form $\zeta_{D^2}(2) = \pi^2 - \pi^4/12$ as a new Paper 28 identity (consistent with $T9$, not previously logged). Sprint provenance: `debug/kcc_sprint_memo.md`.

3. α^5 **Lamb shift residual: scoped, reframed, and confirmed.** Sprint LS-5 (May 2026) scoped the two-loop QED calculation on the framework and found two distinct results, both since confirmed by sprint LS-6a. *First*, the framework is structurally ready for two-loop QED on S^3: the mode-count budget is favorable ($\sim$2,870 terms at $n_{\max} = 20$ from SO(4) selection rules), and the infrastructure pieces exist (`qed_two_loop.py`, `qed_self_energy.py`, the LS-3/LS-4 bound-state Sturmian projection); the work is additive rather than blocked, with an estimate of $\sim$8 follow-on sprints to close the multi-loop ceiling. *Second*, the original test target was misidentified. The genuine two-loop α^5 multi-loop contribution to the $2S$ Lamb

shift is approximately +7.10 MHz (Eides–Grotch–Shelyuto 2001 Table 7.4 cumulative central value), not the $\sim +29$ MHz LS-3 residual. Sprint LS-6a (May 2026) confirmed the reframing: re-deriving the same one-loop physics in the Eides § 3.2 canonical convention identified that the original LS-1 implementation had inadvertently subtracted the Uehling kernel constant $4/15 = (10/9 - 38/45)$ from the canonical 10/9 self-energy coefficient (Uehling already enters separately as a VP contribution). The convention shift from LS-1 to LS-6a is exactly +27.13 MHz at $n = 2$, $Z = 1$, structurally identifiable as $(4/15) \cdot \alpha^3 Z^4/(\pi n^3) \cdot$ (Ha-to-MHz). LS-6a predicts 1052.19 MHz, with residual -5.65 MHz / $-$ 0.534% against experimental 1057.85 MHz — a 5.8× reduction in absolute error vs the LS-1 one-loop ceiling (-32.78 MHz); vs LS-3 (-28.85 MHz) the reduction is 5.1×. Sprint LS-7 (May 2026) made a first pass at the two-loop SE on Dirac-S^3 via iterated CC spectral action and returned two findings. *Structural confirmation*: the two-loop SE prefactor $(\alpha/\pi)^2(Z\alpha)^4/n^3$ emerges with the $1/\pi^2$ tracing to two iterated Schwinger proper-time integrations, exactly as Prediction 1 specifies; weak test because the same prefactor appears in flat-space two-loop QED. *Critical residual reframing*: the -5.65 MHz LS-6a residual is *not primarily multi-loop QED*. The Eides–Grotch–Shelyuto 2001 Table 7.3 (α^5 multi-loop QED proper) totals only $\sim +1.20$ MHz; the remaining $\sim +4.4$ MHz of the residual must be non-loop physics; the itemized non-loop entries are recoil -2.40 MHz, finite nuclear size +0.138 MHz (the 2S value; the retired +1.18 is the 1S shift), hyperfine averaging $\sim +5.0$ MHz. These three sum to $\sim +2.74$ MHz; adding the $\sim +1.20$ MHz multi-loop term gives $\sim +3.94$ MHz for the four itemized contributions against the -5.65 MHz residual — $\sim$ 70% closure, not a match to within the literature precision. (Corrected 2026-08-29: an earlier correction note printed +3.78/+4.98 — sums still computed with the retired +1.18 MHz finite-size entry; the +0.138 MHz 2S value gives +2.74/+3.94, in agreement with Paper 34's Lamb autopsy. Cross-check: $5.65 - 3.94 = 1.71$ MHz reproduces Paper 34's tabulated +1.72 MHz Lamb-autopsy residual — the unattributed remainder IS the autopsy residual.) The earlier LS-5 / LS-6a memos had used Eides Table 7.4 (cumulative α^5 contributions of all kinds) as the multi-loop reference; this was a misreading. Implications: (i) the framework's one-loop closure is even cleaner than originally framed — only $\sim +1.20$ MHz of the residual is missing-multi-loop physics; (ii) the genuine multi-loop test target for Prediction 1 is sharpened to $\sim +1.20$ MHz, not $\sim +5.65$ MHz; (iii) the strong test (sprint LS-8a) is the native derivation of the dimensionless bracket coefficient $C_{2S} = +3.63$ from iterated CC spectral action plus bound-state Sturmian projection, with the literature bracket value used as a check rather than as input. The remaining $\sim +4.4$ MHz of non-loop physics is within scope for the rest-mass projection (Paper 34 § III.14) and Paper 23's nuclear-electronic composed Hamiltonian, but is not itself a test of Prediction 1.

Sprint LS-8a (May 2026) closed the strong-test attempt at C_{2S} extraction with a clean structural verdict. The bare iterated Connes–Chamseddine spectral action on Dirac-S^3 (rainbow plus crossed topologies, full SO(4) vertex selection at four vertices, bound-state Sturmian projection at $n_{\text{ext}} = 1$ for hydrogen $2S$) faithfully reproduces the UV-divergent *integrand* of two-loop QED: the $(\alpha/\pi)^2(Z\alpha)^4/n^3$ prefactor emerges as predicted, the sign is correct at every $n_{\max}$, and the sum diverges as raw $\sim 6.6 \cdot N^{3.43}$—power-law UV divergence inherited from flat-space QED. Two natural regularizations within the GeoVac toolkit fail: subtracting the disconnected $[\Sigma^{1L}]^2$ piece removes <0.1% of the total at every $n_{\max}$ (the divergence sits in the genuine connected diagram), and Drake–Swainson asymptotic subtraction (Paper 34's 13th projection) does not apply to power-law divergences. Finite extraction of C_{2S} requires Z_2 and δm renormalization counterterms which are *not generated* by the bare CC spectral action. This places a clean scope boundary between one-loop closure (achieved at sub-percent in Paper 36 via Drake–Swainson plus the Eides § 3.2 canonical convention) and two-loop closure (requires renormalization machinery the framework does not natively produce). **Refined Prediction 1:** the GeoVac framework controls the π content of the *finite parts* of any QED observable; UV divergences of multi-loop contributions are inherited from the underlying field theory and renormalized by counterterms that the framework does not autonomously generate. This is consistent with and strengthens the original Prediction 1: π enters via temporal integration; renormalization is a separate projection not yet in the Paper 34 catalogue. Sprint provenance: `debug/ls5_two_loop_scoping_memo.md`, `debug/ls6a_eides_convention_memo.md`, `debug/ls7_two_loop_se_memo.md`, and `debug/ls8a_two_loop_sel`

4. **Compactification beyond S^1_β.** We tested only the $S^3 \times S^1_\beta$ compactification. More general temporal identifications (with twists, with curvature, with multiple temporal directions) would give more general π-injection patterns. Whether the prediction (Section 7) extends to all such compactifications, or only to the standard Matsubara case, is open.

10 Conclusion

The bare relativistic Klein-Gordon spectrum on $S^3 \times \mathbb{R}$ is π-free in the algebraic-extension ring $\mathbb{Q}[\sqrt{d_1}, \sqrt{d_2}, \ldots]$ for every rational mass-squared. π enters the spectrum at exactly one step – temporal compactification – through the Matsubara frequency $2\pi k/\beta$. The conformally coupled massless scalar Casimir energy on unit S^3 is the exact rational 1/240; the Stefan-Boltzmann $\pi^2/90$ enters only through the Matsubara sum. The structural reading is that mass is a parameter projection that preserves the algebraic-extension ring of the bare graph; the observation / temporal-window projection is the categorically distinct projection that introduces π. These should be split as separate rows in Paper 34's projection dictionary.

The framework-wide prediction is that a GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter promoted from the discrete graph spectrum. Existing GeoVac observables uniformly confirm this prediction: Pauli counts, magic numbers, hydrogenic eigenvalues, Bargmann–Segal lattice, and graph-native QED matrix elements are all π-free; vacuum polarization, β-function, and Stefan-Boltzmann all contain π exactly because their evaluation includes a continuous Schwinger-proper-time or Matsubara integration.

With the addition of Sprint TX-B's five-observable falsification panel (§ 7.3), the prediction has been tested on a cumulative 208/208 individual checks across the KG and TX-B sprints with no exceptions, and graduates from falsifiable conjecture to load-bearing principle of the framework (Remark 4). TX-B also produced a new clean closed form: the spatial Casimir of the 3D HO on the Bargmann–Segal S^5 lattice is $E_{\mathrm{Cas}}^{\mathrm{HO},S^5} = -17/3840$ (exact rational), structurally a sibling of the scalar S^3 Casimir (1/240, KG-3) and the Dirac S^3 Casimir (17/480, KG-5). The shared mechanism is half-integer Hurwitz at shift $a = 3/2$, with Bernoulli polynomials at 3/2 rational — the universal driver of rationality across all three Casimir sectors. The Coulomb/HO/Dirac asymmetry of Paper 24 (operator-order, calibration π, Wilson gauge with matter) does not extend to the Casimir transcendental class.

The sharpest open question this paper does not resolve is the Paper 2 $K = \pi(B+F-\Delta) \approx 1/\alpha$ coincidence: the explicit π on the right-hand side is a candidate counter-example to the prediction, but only if the projections that produce K are genuinely discrete-graph quantities and not single-projection spectral-action-type statements in disguise. Sprint K-CC has since *ruled out* the single-projection reading at the framework's current resolution (no single Chamseddine–Connes heat-kernel expansion contains B, F and Δ as terms); the live readings are the revised multi-projection one and the coincidence one, per WH5. The open step is therefore to discriminate *those two*, a structural follow-on to Sprint A (2026-04-18) and the WH1 Marcolli–van Suijlekom upgrade. (Updated 2026-08-28: the earlier wording proposed as a next step a question this paper's own Section 9 reports closed.)

References

[1] GeoVac Project, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," Paper 0 (2026).

[2] GeoVac Project, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," Paper 7 (2025).

[3] GeoVac Project, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," Paper 14 (2025).

[4] GeoVac Project, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," Paper 18 (2026).

[5] GeoVac Project, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," Paper 22 (2026).

[6] GeoVac Project, "Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice," Paper 23 (2026).

[7] GeoVac Project, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," Paper 24 (2026).

[8] GeoVac Project, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," Paper 28 (2026).

[9] GeoVac Project, "The Universal/Coulomb Partition of the GeoVac Spectral Triple," Paper 31 (2026).

[10] GeoVac Project, "The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage," Paper 32 (2026).

[11] GeoVac Project, "The 1+6+1 Partition of QED Selection Rules on S^3," Paper 33 (2026).

[12] GeoVac Project, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," Paper 34 (2026).

[13] GeoVac Project, "Sprint KG Memo: Klein-Gordon on $S^3 \times \mathbb{R}$ – where does π enter?" `debug/kg_sprint_memo.md` (May 2026).

[14] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space*, Cambridge University Press, 1982.

[15] A. A. Bytsenko, G. Cognola, E. Elizalde, V. Moretti, and S. Zerbini, *Analytic Aspects of Quantum Fields*, World Scientific, 2003 (§ 4.5).

[16] J. S. Dowker and R. Critchley, "Effective Lagrangian and energy momentum tensor in de Sitter space," *Phys. Rev. D* **13**, 3224 (1976).

[17] L. H. Ford, "Quantum Vacuum Energy in General Relativity," *Phys. Rev. D* **11**, 3370 (1975).

[18] R. Camporesi and A. Higuchi, "On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces," *J. Geom. Phys.* **20**, 1 (1996).

[19] V. Morinelli, "A geometric perspective on Algebraic Quantum Field Theory," arXiv:2412.20410 (2024).

[20] J. E. Tener, "The Bisognano-Wichmann property for non-unitary Wightman conformal field theories," arXiv:2506.10625 (2025).

[21] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17**, 303–321 (1976).

[22] G. L. Sewell, "Quantum fields on manifolds: PCT and gravitationally induced thermal states," *Annals of Physics* **141**, 201–224 (1982).

[23] J. B. Hartle and S. W. Hawking, "Path-integral derivation of black-hole radiance," *Phys. Rev. D* **13**, 2188–2203 (1976).

[24] W. G. Unruh, "Notes on black-hole evaporation," *Phys. Rev. D* **14**, 870–892 (1976).